

Beam Current Limits

USPAS - Beam Physics with Intense Space Charge

Austin Hoover Steven Lund Daniel Winklehner John Barnard

July 2026

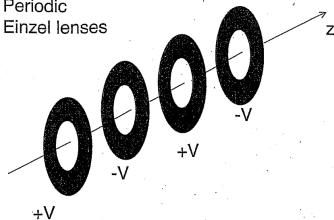
Outline I

1. Beam current limits in axisymmetric focusing channels
2. Beam current limits in quadrupolar focusing channels

Outline

1. Beam current limits in axisymmetric focusing channels
2. Beam current limits in quadrupolar focusing channels

Periodic
Einzel lenses



PERIODIC SOLENOIDS

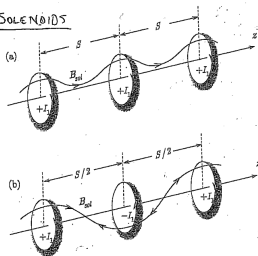


Figure 3.2. Schematic of magnet sets producing a periodic focusing solenoidal field with axial periodicity length S . In Fig. 3.2 (a), successive coils are spaced by S and have the same current polarity $+I_1, +I_1, \dots$. In Fig. 3.2 (b), successive coils are spaced by $S/2$ and have alternating current polarities $+I_1, -I_1, +I_1, \dots$.

(FIGURE FROM
DAVIDSON & QIN,
2003) P. 55
"PHYSICS OF
INTENSE CHARGED
PARTICLE BEAMS
IN HIGH ENERGY
ACCELERATORS"

1.1 Radial envelope equation

$$\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta \rangle}{\gamma m\beta c}\right)^2\frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{2\tilde{r}} = 0 \quad (1)$$

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + q r A_\theta, \quad (2)$$

$$\varepsilon_r \equiv \sqrt{\langle r^2 \rangle \langle r'^2 \rangle - \langle r r' \rangle^2 + \langle r^2 \rangle \langle r^2 \theta'^2 \rangle - \langle r^2 \theta' \rangle^2} \quad (3)$$

$$Q \equiv \frac{1}{mc^2 \beta^2 \gamma^3} \frac{q \langle \lambda(r) \rangle}{2\pi \epsilon_0} \quad (4)$$

1.2 Beam current limit in long solenoid

- Long solenoid: $\mathbf{B} = B_z \hat{z}$.
- No acceleration $\gamma' = \gamma'' = 0$.
- Set $p_\theta = 0$ and $\epsilon_r = 0$ to minimizing “repelling” forces.

$$\tilde{r}'' + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \quad (5)$$

Maximum beam perveance occurs at equilibrium, when $\tilde{r}' = \tilde{r}'' = 0$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma\beta c} \right)^2 \quad (6)$$

1.3 Derivation from force balance at low energies

- Particle moves with angular velocity $\theta = \omega r$.
- Particle sees linear focusing force from solenoid field $B_z = \omega_c m/q$.
- Within beam envelope, particle sees linear defocusing force from space charge.

$$\underbrace{m\omega^2 r}_{\text{centrifugal}} + \underbrace{\frac{Qm\beta^2 c^2}{2\tilde{r}^2} r}_{\text{centrifugal}} = \underbrace{qv_\theta B_z}_{\text{magnetic}} \quad (7)$$

$$\omega^2 r + \frac{Q\beta^2 c^2}{2\tilde{r}^2} r = \omega_c \omega r \quad (8)$$

$$Q(\omega) = \frac{2\tilde{r}^2}{\beta^2 c^2} (\omega\omega_c - \omega^2) \quad (9)$$

$$\omega^* = \max_{\omega} Q(\omega) = \frac{\omega_c}{2} \quad (10)$$

$$Q(\omega^*) = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\beta c} \right)^2 \quad (11)$$

Equivalent to result from envelope equations when $\gamma^2 \rightarrow 1$:

$$Q = \frac{1}{2} \left(\frac{\omega_c \tilde{r}}{\gamma \beta c} \right)^2 \quad (12)$$

1.4 Angular kick at solenoid entrance

Approximate solenoid field as step function:

$$B_z(z) = B_0 [\Theta(z) + \Theta(L - z) - 1] = \begin{cases} 0 & \text{if } z < 0 \\ B_0 & \text{if } 0 \leq z \leq L \\ 0 & \text{if } z > L \end{cases} \quad (13)$$

$$\frac{\partial B_z}{\partial z} = B_0 [\delta(z) - \delta(L - z)] \quad (14)$$

Use earlier result to relate longitudinal derivative to radial magnetic field:

$$B_r(r, z) \approx -\frac{r}{2} \frac{\partial B_z}{\partial z} = \frac{rB_0}{2} [\delta(z) - \delta(L - z)] \quad (15)$$

Compute change in θ component of momentum at $z = \epsilon$, just after solenoid entrance:

$$\frac{d}{dt} p_\theta^*(r) = qv_z B_r \quad (16)$$

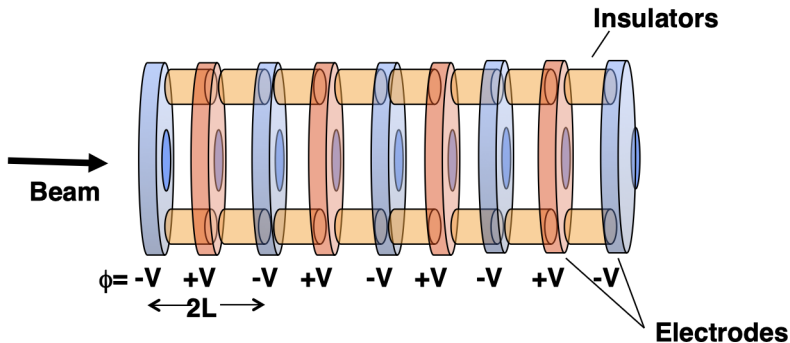
$$\Delta p_\theta^*(r) = q \int_{-\infty}^{\epsilon} B_r(r, z) dz \quad (17)$$

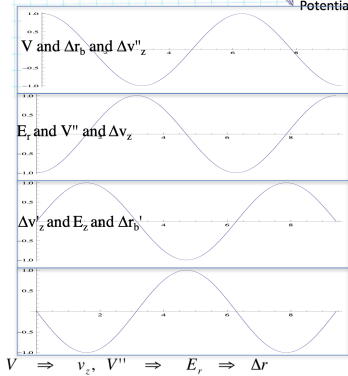
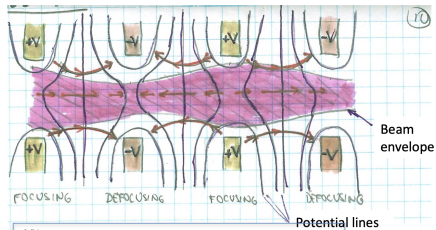
$$= \frac{qrB_0}{2} \int_{-\infty}^{\epsilon} [\delta(z) - \delta(L - z)] dz \quad (18)$$

$$= \frac{qrB_0}{2} \quad (19)$$

$$v_\theta = \frac{rqB_0}{2m} = \frac{r\omega_c}{2} \quad (20)$$

1.5 Beam current limits in periodic Einzel lens





Ignore centrifugal term, emittance term, and solenoid term in radial envelope equation.

$$\tilde{r}'' + \frac{\gamma'}{\gamma\beta^2}\tilde{r}' + \frac{\gamma''}{2\gamma\beta^2} - \frac{Q}{2\tilde{r}} = 0 \quad (21)$$

Assume non-relativistic beam: $\beta \ll 1$, $\gamma \approx 1$:

$$\gamma' = \beta\beta'\gamma^3 \approx \beta\beta' \quad (22)$$

$$\gamma'' \approx \beta'\beta' + \beta\beta'' \quad (23)$$

$$\tilde{r}'' + \frac{\beta'}{\beta}\tilde{r}' + \frac{1}{2} \left[\frac{\beta'^2}{\beta^2} + \frac{\beta''}{\beta} \right] \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \quad (24)$$

Define new variable $R = \sqrt{\beta/\beta_0}\tilde{r}$, where β_0 is the initial velocity.

$$\tilde{r}' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R' - \frac{1}{2} \left(\frac{\beta}{\beta_0}\right)^{-3/2} \left(\frac{\beta'}{\beta}\right) R \quad (25)$$

$$\tilde{r}'' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R'' - \left(\frac{\beta}{\beta_0}\right)^{-3/2} \left(\frac{\beta'}{\beta_0}\right) R' + \frac{3}{4} \left(\frac{\beta}{\beta_0}\right)^{-5/2} \left(\frac{\beta'}{\beta_0}\right)^2 R - \frac{1}{2} \left(\frac{\beta}{\beta_0}\right)^{-3/2} \left(\frac{\beta'}{\beta}\right) \left(\frac{\beta''}{\beta}\right) R \quad (26)$$

$$R'' + \frac{3}{4} \left(\frac{\beta'}{\beta}\right)^2 R - \frac{1}{2} \frac{\beta}{\beta_0} \frac{Q}{R} = 0.$$

(27)

Approximate potential as $\phi(r, z) = \phi_0 \cos(\pi z/L)$. From conservation of energy:

$$\Delta \left(\frac{1}{2} m v^2 \right) = -\Delta \left(\frac{q\phi}{m} \right) \quad (28)$$

$$\frac{1}{2} m (v^2 - v_0^2) = -q\phi_0 \cos \left(\frac{\pi z}{L} \right) \quad (29)$$

$$v^2 = v_0^2 - \frac{2q\phi_0}{m} \cos \left(\frac{\pi z}{L} \right) \quad (30)$$

Use this to rewrite $(\beta'/\beta)^2$:

$$v' = \frac{q\phi_0}{mv} \left(\frac{\pi}{L} \right) \sin \left(\frac{\pi z}{L} \right) \quad (31)$$

$$\frac{v'}{v} = \frac{q\phi_0}{mv^2} \left(\frac{\pi}{L} \right) \sin \left(\frac{\pi z}{L} \right) \quad (32)$$

$$\left(\frac{\beta'}{\beta} \right)^2 = \left(\frac{q\phi_0}{mv} \right)^2 \left(\frac{\pi}{L} \right)^2 \sin^2 \left(\frac{\pi z}{L} \right) \quad (33)$$

Assume $v_0^2 \gg \frac{2q\phi_0}{m}$, so that $v \approx v_0$.

$$\left(\frac{\beta'}{\beta}\right)^2 \approx \left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \sin^2\left(\frac{\pi z}{L}\right) \quad (34)$$

For equilibrium ($R'' = R' = 0$), look at averages over z :

$$\overline{\left(\frac{\beta'}{\beta}\right)^2} = \frac{1}{2} \left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \quad (35)$$

$$\overline{\left(\frac{\beta}{\beta_0}\right)} = 0 \quad (36)$$

$$\overline{R} = \overline{\left(\frac{\beta}{\beta_0}\right)^{1/2}} \tilde{r} = \tilde{r} \quad (37)$$

Inserting these terms in the envelope equation with $R'' = 0$ gives the maximum beam perveance:

$$\boxed{Q_{max} = \frac{3\pi^2}{4} \left(\frac{q\phi_0}{m\beta_0^2 c^2}\right)^2 \left(\frac{\tilde{r}}{L}\right)^2} \quad (38)$$

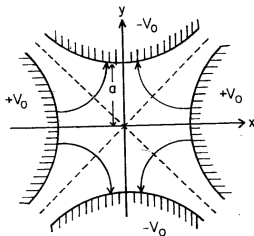
Outline

1. Beam current limits in axisymmetric focusing channels
2. Beam current limits in quadrupolar focusing channels

FROM
REISEL, p.112

$$E_x = -E'_x$$

$$E_y = E'_y$$



$$F_x = -qE'_x$$

$$F_y = qE'_y$$

ELECTROSTATIC
QUADS

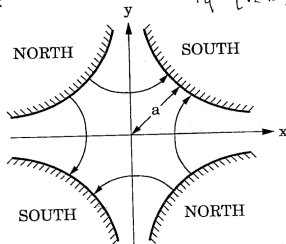
Figure 3.15. Electrodes and force lines in an electrostatic quadrupole.

$$B_x = B'_y$$

$$B_y = B'_x$$

$$F_x = -qV_z B'_x$$

$$F_y = qV_z B'_y$$



MAGNETIC
QUADS

2.1 RMS envelope equations

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x^2}{\tilde{x}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0 \quad (39)$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0 \quad (40)$$

$$Q = \frac{q\lambda}{2\pi\epsilon_0 mc^2 \beta^2 \gamma^3} \quad (41)$$

$$\varepsilon_x(s) = \sqrt{\langle xx \rangle \langle x'x' \rangle - \langle xx' \rangle \langle xx' \rangle} \quad (42)$$

$$\varepsilon_y(s) = \sqrt{\langle yy \rangle \langle y'y' \rangle - \langle yy' \rangle \langle yy' \rangle} \quad (43)$$

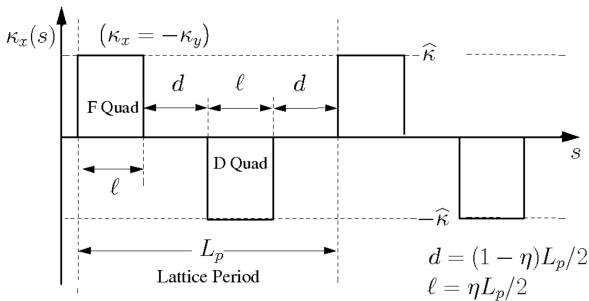
$$\kappa_x(s) = + \frac{qG}{\gamma\beta mc} \quad (44)$$

$$\kappa_y(s) = - \frac{qG}{\gamma\beta mc} \quad (45)$$

Let $\kappa_x(s) = f(s)\kappa$ and $\kappa_y(s) = -\kappa_x(s)$. Ignoring acceleration ($\gamma' = 0$):

$$\tilde{x}'' + \kappa f(s)\tilde{x} - \frac{\varepsilon_x^2}{\tilde{x}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0 \quad (46)$$

$$\tilde{y}'' - \kappa f(s)\tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0 \quad (47)$$



$\eta = \text{Occupancy} \in (0, 1]$

$$f(s) = \begin{cases} +1 & \text{if } 0 \leq s < L\eta/2 \\ -1 & \text{if } L - \eta L/2 \leq s < L + L\eta/2 \\ +1 & \text{if } 2L - L\eta/2 \leq s < 2L \end{cases} \quad (48)$$

To derive the maximum beam current, approximate $f(s)$, $\tilde{x}(s)$, and $\tilde{y}(s)$ as sinusoidal functions at the lattice frequency. Then plug into envelope equation, ignore fast-oscillating terms, and solve for equilibrium.

Start with the focusing strength $f(s)$:

$$f(s) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi s}{L}\right) \quad (49)$$

$$a_n = \frac{1}{L} \int_0^{2L} f(s) \cos\left(\frac{n\pi s}{L}\right) ds \quad (50)$$

$$f(s) \approx a_1 \cos\left(\frac{\pi s}{L}\right) = \frac{4}{\pi} \sin\left(\frac{\pi\eta}{2}\right) \cos\left(\frac{\pi s}{L}\right) \quad (51)$$

Approximate rms envelope evolution as:

$$\tilde{x}(s) = \frac{\tilde{r}_0}{\sqrt{2}} \left[1 + \delta \cos \left(\frac{\pi s}{L} \right) \right] \quad (52)$$

$$\tilde{y}(s) = \frac{\tilde{r}_0}{\sqrt{2}} \left[1 - \delta \cos \left(\frac{\pi s}{L} \right) \right] \quad (53)$$

Now evaluate terms in rms envelope equations: $\tilde{x}''$, $f(s)\tilde{x}$, etc.

$$\tilde{x}'' = -\frac{\tilde{r}_0}{\sqrt{2}} \delta \frac{\pi^2}{L^2} \cos \left(\frac{\pi s}{L} \right) \quad (54)$$

$$\tilde{y}'' = +\frac{\tilde{r}_0}{\sqrt{2}} \delta \frac{\pi^2}{L^2} \cos \left(\frac{\pi s}{L} \right) \quad (55)$$

$$f(s)\tilde{x}(s) = \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \cos\left(\frac{\pi s}{L}\right) \left[1 + \delta \cos\left(\frac{\pi s}{L}\right)\right], \quad (56)$$

$$= \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) + \delta \cos^2\left(\frac{\pi s}{L}\right)\right]. \quad (57)$$

Ignore fast oscillating component of $\cos^2(\pi s/L)$:

$$\cos^2\left(\frac{\pi s}{L}\right) = \frac{1}{2} + \frac{1}{2} \cos\left(\frac{2\pi s}{L}\right) \approx \frac{1}{2}. \quad (58)$$

This gives:

$$f(s)\tilde{x}(s) \approx \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) + \frac{\delta}{2}\right], \quad (59)$$

$$f(s)\tilde{y}(s) \approx \frac{4\tilde{r}_0}{\sqrt{2}\pi} \sin\left(\frac{\pi\eta}{2}\right) \left[\cos\left(\frac{\pi s}{L}\right) - \frac{\delta}{2}\right]. \quad (60)$$

Plugging these approximations into rms envelope equations:

$$-\left[\delta \frac{\pi^2}{L^2} - \frac{4\kappa}{\pi} \sin\left(\frac{\eta\pi}{2}\right)\right] \cos\left(\frac{\pi s}{L}\right) + \frac{2\kappa\delta}{\pi} \sin\left(\frac{\eta\pi}{2}\right) = \frac{Q}{2\tilde{r}_0^2}, \quad (61)$$

$$+\left[\delta \frac{\pi^2}{L^2} - \frac{4\kappa}{\pi} \sin\left(\frac{\eta\pi}{2}\right)\right] \cos\left(\frac{\pi s}{L}\right) + \frac{2\kappa\delta}{\pi} \sin\left(\frac{\eta\pi}{2}\right) = \frac{Q}{2\tilde{r}_0^2}. \quad (62)$$

Solve for δ and Q :

$$\delta = \frac{4\kappa L^2}{\pi^3} \sin\left(\frac{\eta\pi}{2}\right) \quad (63)$$

$$Q = \frac{4\kappa\delta}{\pi} \sin\left(\frac{\eta\pi}{2}\right) \tilde{r}_0^2 \quad (64)$$

$$\boxed{Q_{max} \approx \left(\frac{2\kappa L\eta}{\pi}\right)^2 \left(\frac{\sin(\eta\pi/2)}{(\eta\pi/2)}\right)^2 \tilde{r}_0^2} \quad (65)$$

2.2 Continuous focusing

$$\tilde{x}'' + \omega_0^2 \tilde{x} - \frac{\varepsilon^2}{\tilde{x}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0, \quad (66)$$

$$\tilde{y}'' + \omega_0^2 \tilde{y} - \frac{\varepsilon_y^2}{\tilde{y}^3} - \frac{Q}{2(\tilde{x} + \tilde{y})} = 0. \quad (67)$$

At equilibrium ($\tilde{x}'' = \tilde{y}'' = 0$), the defocusing space charge term balances the external focusing term. This occurs at the equilibrium rms radius $\tilde{r}_0 = \sqrt{2}\tilde{x} = \sqrt{2}\tilde{y}$.

$$Q_{max} = 2\omega_0^2 \tilde{r}_0^2 \quad (68)$$

Comparing to Eq. (65) gives the oscillation frequency ω_0^2 for the smooth-focusing approximation of the FODO channel:

$$\omega_0^2 = \frac{2\kappa^2 L^2 \eta^2}{\pi^2} \left(\frac{\sin(\eta\pi/2)}{(\eta\pi/2)} \right)^2 \quad (69)$$

Define the phase advance $\sigma_0 = \omega_0/(2L)$:

[...]

2.3 Summary

$$Q_{max} = \dots$$

$$Q_{max} = \dots$$

$$Q_{max} = \dots$$